7.1.2. According to the Fundamental Mechanism Used

1. Negative Resistance Oscillators
2. Feedback Oscillators

Negative resistance oscillator has the negative resistance amplifying device to neutralize the positive resistance of the oscillator. Feedback oscillator uses positive feedback in the feedback amplifiers to satisfy the Barkhausen criterion

7.1.3. According to the Frequency Generated

1. Audio frequency (AF) oscillator: 20Hz to 20 kHz
2. Radio frequency (RF) oscillator 20 kHz-30 MHz
3. Very high frequency (VHF) oscillator: 30 MHz-300 MHz
4. Ultra high frequency (UHF) oscillator: 300 MHz-3 GHz
5. Microwave frequency oscillator: above 3GHz

7.2. TYPES OF SINUSOIDAL OSCILLATION

Sinusoidal electrical oscillation can be of two types

1. Damped oscillation
2. Undamped oscillation

7.2.1. Damped Oscillation

Figure 7.2(a) shows the damped oscillation. In this type of oscillators, during each oscillation, some energy is lost due to electrical losses (I^2R).The amplitude of the oscillation reduced to zero, when no compensating arrangement for the electrical losses is provided.

7.2.2. Undamped Oscillation

Figure 7.2(b) shows the undamped oscillations. In these types, the amplitude of each oscillation remains constant with time. Although the electrical system in which these oscillations are being generated has losses, but now right amount of energy is being supplied to overcome these losses. Therefore, the generated wave remains constant. It is also called as sustained oscillations. These continuous waves are produced by electronic oscillator circuits for utilizing in various electronic equipment.

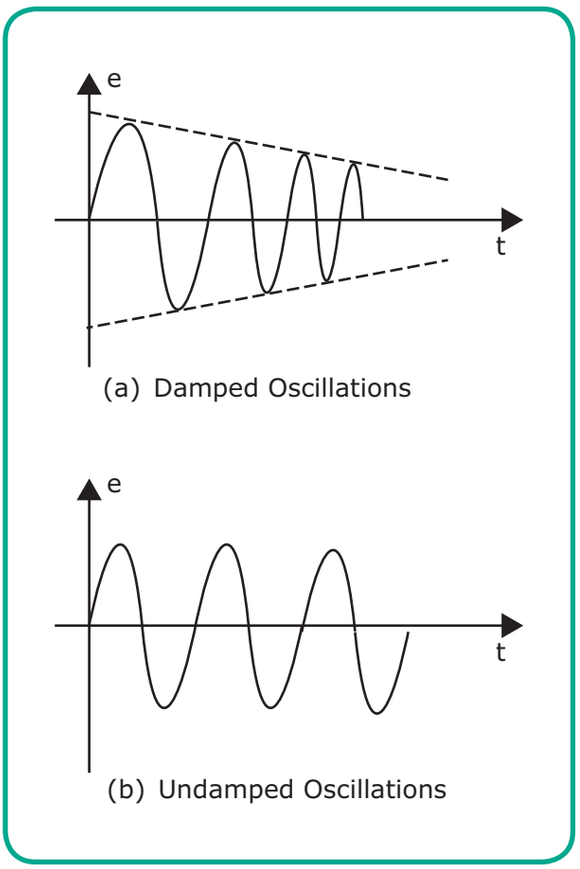


Figure 7.2 **Damped and Undamped Oscillations**

ELEC_EQUIP_Chapter07.indd 150 11/10/2020 11:46:11 AM